

SAMPLING RATE CONVERSION

Finding the Computational Error

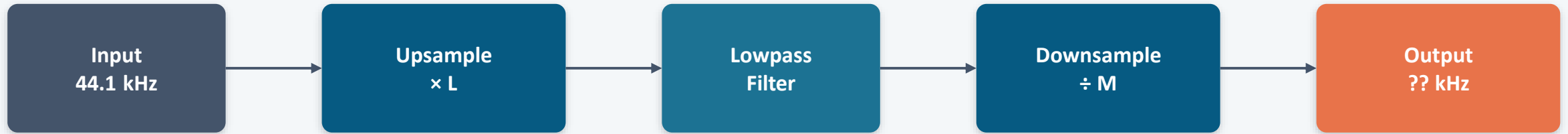
Why 44.1 kHz $\rightarrow$ 48 kHz conversion produced 40.5 kHz instead — and how to fix it

$$L = 147 \rightarrow M = 160$$

as given in the problem

THE PROBLEM

A multirate converter that missed its target



Given values

Input rate f_{in}	44.1 kHz
Target output rate f_{out}	48 kHz
Interpolation factor L	147
Decimation factor M	160
Actual measured output	40.5 kHz X

GOVERNING RELATION

$$f_{out} = f_{in} \times (L / M)$$

The signal is first **upsampled by L** (interpolation), lowpass filtered, then **downsampled by M** (decimation).

DIAGNOSIS

L and M were assigned to the wrong roles

The reduced fraction of 48000 / 44100 is 160 / 147. That fraction fixes **which integer multiplies and which divides** — it does not fix which one is called “L” and which is called “M.” The problem swapped them: it upsampled by the **smaller** number and downsampled by the **larger** one — the reverse of what's needed to raise the rate.

AS GIVEN (WRONG)

Upsample by L **147**

Downsample by M **160**

$$f_{\text{out}} = 44.1 \times (147 / 160)$$

$$= 40.516875 \text{ kHz} \approx 40.5 \text{ kHz}$$

Matches the faulty measured output — confirms the swap.

CORRECTED

Upsample by L **160**

Downsample by M **147**

$$f_{\text{out}} = 44.1 \times (160 / 147)$$

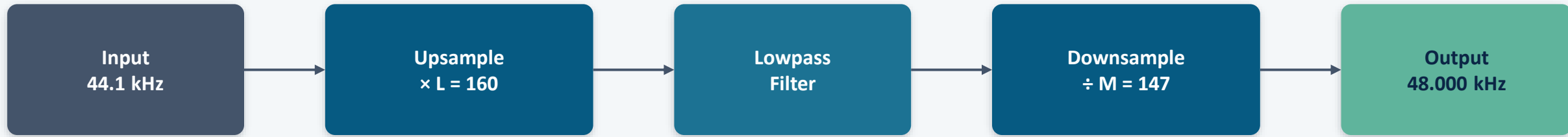
$$= 48.000 \text{ kHz exactly}$$

$$44100 / 147 = 300 \text{ (integer)} \rightarrow 300 \times 160 = 48000.$$

WHY THE ORDER MATTERS

Upsample by the larger factor, downsample by the smaller

Since 48 kHz > 44.1 kHz, the rate must increase overall, so the multiplying factor **L** must be the larger integer (160) and the dividing factor **M** the smaller one (147). Swapping them, as in the original setup, silently inverts the ratio and lowers the rate instead.



STEP-BY-STEP VERIFICATION

$$\text{GCD}(44100, 48000) = 300$$

$$44100 \div 300 = 147 \quad \text{and} \quad 48000 \div 300 = 160$$

$$\Rightarrow L = 160 \quad (\text{multiply / upsample}), \quad M = 147 \quad (\text{divide / downsample})$$

$$f_{\text{out}} = 44100 \times (160 / 147) = 300 \times 160 = 48,000 \text{ Hz} = 48 \text{ kHz} \checkmark$$

CONCLUSION

The error was a swapped L and M — not a bad ratio

L = 147, M = 160

Original (incorrect) assignment

L = 160, M = 147

Corrected assignment

48.000 kHz

Correct output rate

Takeaway: In L/M rate conversion, the reduced fraction fixes the ratio, but **L must be the interpolation (upsampling) factor and M the decimation (downsampling) factor**. Reversing them inverts the ratio and produces an incorrect — and in this case lower — output rate.